BRAC UNIVERSITY
Inspiring Excellence

# Remedial Course in Mathematics
# Final Examination
# Total - 40 Marks

(You need to answer **ANY FOUR** questions, and return the question with answer script.)

(Time duration: 1 hr 30 min)

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1. (a) Sketch step-by-step the graph of

$$y = 1 + 5\cos\left(2x - \frac{\pi}{2}\right).$$

Mark at least one full period.

(b) On the same set of axes, sketch the graphs of

$$y = |\sin x| \quad \text{and} \quad y = \cos x$$

for $0 \leq x < 2\pi$. Clearly mark all points of intersection. Hence, solve the equation

$$|\sin x| = \cos x$$

for $0 \leq x < 2\pi$.

(5+5 Marks)

2. Given the system of equations:

$$\begin{aligned} x &= 1 \\ 2y + 3z &= 2 \\ y + z &= 3 \end{aligned}$$

(a) Write the system in matrix form, $A\vec{x} = \vec{b}$.

(b) Determine whether the system is solvable or not.

(c) If the system is solvable find the solution using any method.

(2+4+4 Marks)

3. (a) Find the fourth root of $z = -1 - i$. And sketch the root in the complex plane.

(b) Write the complex number

$$z = -1 + i\sqrt{3}$$

in the polar form

$$z = r(\cos\theta + i\sin\theta)$$

Hence, compute $z^{100}$.

(6+4 Marks)

4. (a) Eight friends go out for dinner. How many ways are there to sit them around a round table? Rotations of a sitting arrangement are considered the same, but a reflection is considered different.

(b) If the reflections are also considered the same, how many ways are there to sit them around a round table?

(c) A team of four has to be selected from 6 boys and 4 girls. How many different ways can a team be selected if at least one boy must be in the team?

(d) How many different diagonals does a 12-sided polygon have?

(e) How many triangles can be formed using 10 points in a plane, out of which 4 are collinear?

(2+2+2+2+2 Marks)

5. A **Riemann sum** approximates this area using rectangles:

$$\sum_{i=1}^{n} f(x_i^*)\,\Delta x \quad \text{where} \quad \Delta x = \frac{b-a}{n}.$$

As $n \to \infty$, rectangles become very thin and the approximation becomes exact.

(a) Find the value of the definite integral using formula

$$\int_0^1 x^2\,dx.$$

(b) We want to approximate the same integral. Take $n = 4$ equal subintervals and use **left endpoints** to compute the Riemann sum.

(c) Consider the differential equation

$$\frac{dy}{dx} = x.$$

Find the slope field and the solution family.

(2+5+3 Marks)

6. (a)

$$f(x) = \begin{cases} x^2 + k, & x \leq 2 \\ 4x - 3, & x > 2 \end{cases}$$

Find $k$ such that $\lim_{x \to 2} f(x)$ exists.

(b) Evaluate:

$$\lim_{x \to 4} \frac{\sqrt{x} - 2}{x - 4}.$$

(6+4 Marks)

7. (a) Identify and sketch:

$$|z - 1| + |z - 1| = 4$$

where $z = x + iy$.

(b)

$$\begin{aligned} 2x + 4y &= 6 \\ x + 2y &= 3 \end{aligned}$$

Can you solve the system of equations? Give reasons based on determinants.

(c) Solve the system using $X = A^{-1}b$:

$$\underbrace{\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} x \\ y \end{pmatrix}}_{\vec{x}} = \underbrace{\begin{pmatrix} 5 \\ 11 \end{pmatrix}}_{\vec{b}}.$$

(6+2+2 Marks)

**In remembrance of Sharif Osman Bin Hadi, whose courage and sacrifice will shape our nation.**